Real-time Multiparty Disembodied-LLM Dialogue System for Education

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Conversational Agents Course – Final Report
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Abstract

This project explores vicarious learning within a real-time spoken dialog framework. We present a multi-agent system where a Teacher agent delivers educational content while a Student agent acts as a synthetic peer, interjecting with clarifying questions to draw out more detailed explanations. Unlike traditional text-based tutors, our system utilizes a low-latency speech interface, integrating Whisper for STT and Chatterbox for TTS, to foster a natural, triadic "fictive dialog" environment. By allowing human learners to overhear and join this agentic interaction, we aim to reduce cognitive load and improve knowledge retention. Evaluation with human participants suggests that the presence of an autonomous student agent enhances engagement and simulates a dynamic classroom experience.

1 Introduction

1.1 Problem Statement and Motivation

While LLM-based educational agents offer high linguistic fluency, most existing systems rely on dyadic (one-on-one) text interactions that fail to capture the social dynamics of classroom learning. A key pedagogical gap lies in the absence of vicarious learning, the process where students gain knowledge by observing interactions between others. Passive learners often struggle to identify their own knowledge gaps, a problem exacerbated by the cognitive load of managing a dialogue. By moving away from text-only interfaces toward a triadic, real-time spoken "fictive dialogue", we can simulate a more natural and effective learning environment that triggers deeper reflection through agentic exchanges.

1.2 Research Aims and Objectives

The primary aim of this project is to demonstrate how a multi-agent architecture can foster vicarious learning through autonomous student-teacher interactions. The specific objectives are:

1. To develop a Teacher Agent capable of delivering structured, pedagogically sound content via a spoken channel.
2. To implement an autonomous Student Agent that mimics human curiosity by interjecting with clarifying questions.
3. To integrate low-latency speech technologies (Whisper STT and Chatterbox TTS) to manage a fluid triadic interaction.
4. To evaluate the impact of a "synthetic peer" on user engagement and conceptual retention through controlled human testing.

1.3 Experimental overview

To demonstrate the effectiveness of our hypothesis, we conducted a user study comparing two pedagogical conditions: a standard dyadic interaction (Teacher-Human) and our proposed triadic fictive dialog (Teacher-Student-Human).

Participants are tasked with learning a specific topic through a spoken session in real-time. We measure learning gains through pre- and post-interaction assessments, while qualitative engagement is captured through subjective questionnaires. This setup allows us to isolate the impact of the Student Agent's interjections on the user's understanding and overall experience.

2 Previous Work

2.1 Dialogue, Vicarious Learning, and Educational Conversational Agents

For nearly two decades, spoken dialogue systems (SDS) have been investigated in educational contexts, particularly for intelligent tutoring (Litman and Silliman, 2004; Schultz et al., 2004) and language learning

(Seneff et al., 2007; Bibauw et al., 2019), and more recently as conversational agents powered by large language models (Sydorenko et al., 2025). Dialogue has been demonstrated to be a more effective modality than plain text for knowledge retention (Ginns et al., 2013), and merely *overhearing* dialogue between agents has itself been shown to support learning better than listening to a monologue (Driscoll et al., 2003).

This phenomenon, coined vicarious learning, has been investigated through "fictive dialogues": semi-scripted exchanges between virtual agents generated from a source text, where the learner is a passive listener rather than an active participant (Piwek et al., 2007). Aylett et al. (2024) extend this concept to a classroom language-learning Educational Conversational Agent (ECA) that alternates between teacher and peer roles. They find that more scripted agent and human-confederate exchanges improve comprehension, whereas direct agent-to-learner questioning degrades it. The multi-party configurations needed for designing unscripted agentic conversations remain a known open challenge for current systems (Aylett and Romeo, 2023; Skantze, 2021), primarily because realtime end-of-turn prediction, speaker detection and interruption handling all become critical in a non-deterministic dialogue.

2.2 LLM-based and Multi-Agent Tutors

A recent systematic review of 36 educational chatbots finds that teaching agents and peer agents together dominate role assignments (Kuhail et al., 2023). This decomposition maps directly onto the agentic teacher and question-asking student configuration the present system adopts. However, empirical evidence on the pedagogical effects of the LLM-based variants of such systems is only beginning to accumulate. Spangenberger et al. (2026) evaluate chat-based conversation with an LLM in education for sustainable development, and find that LLM chat engages both affective and cognitive processes relevant to learning, and results in knowledge gains comparable to reading the same material as text. This establishes LLM chat as a viable teaching medium, but does fall short in larger, structured tasks such as robust curricular sequencing and student modelling. This underpinned the learning gains reported for earlier conversational intelligent tutoring systems such as AutoTutor (Graesser, 2016).

A recent architectural response to recover the teaching structure, is by composing several specialised LLM agents rather than relying on a single chatbot or a scripted dialogue policy. The WikiHowAgent system of Pei et al. (2025) implements teacher, student, interaction-manager and evaluator agents at scale. Shi et al. (2025) apply a similar teacher-student-evaluator framework (EducationQ) to benchmark teaching ability across 14 LLMs. Both systems, however, place no human in the interactive loop, typically the human enters only as a post-hoc judge. Our work attempts to close this gap using a human learner, embedded in realtime dialogue rather than generated text, that follows a similar student-teacher-evaluator role decompostion.

2.3 Speech Interfaces for Educational Agents

The few LLM-based ECAs that do operate in through speech typically pair their dialogue manager with a cloud-based speech I/O pipeline. Aylett et al. (2024), for instance, integrate Deepgram's hosted ASR and TTS synthesis into a RASA dialogue agent. Sydorenko et al. (2025) identify recognition error and end-to-end latency as the two persistent bottlenecks for spoken educational agents. This is a concern espeacially relevant to realtime turn-taking (Skantze, 2021). The present system adopts the same two-component STT/TTS split but runs both locally (using Whisper for STT, and Chatterbox for TTS). By doing this, we trade on-device compute for lower latency and no external dependency in the realtime loop, which addresses the key issues (Sydorenko et al., 2025) for fluent spoken interaction.

2.4 Summary

Our system sits at the intersection of three research fields: the dialogue-based intelligent tutoring field (Litman and Silliman, 2004; Schultz et al., 2004) with its spoken, pedagogically-structured interaction. The recent LLM-based tutor research (Spangenberger et al., 2026; Pei et al., 2025), which contributes fluent generation and scalable multi-agent role decomposition but operates in text-only, typically restricted to one-to-one agent dialogues or monologues. Lastly, the vicarious dialogue paradigm (Driscoll et al., 2003; Piwek et al., 2007; Aylett et al., 2024), which shows that overhearing pedagogically structured dialogue between agents improves learning. The system's central design move is to substitute the scripted human confederate of Aylett et al. (2024) with a second autonomous LLM agent. This serves as a novel combination of the teacher-learner agent decomposition of Pei et al. (2025) with the overhearing-learner stance of the fictive-dialogue tradition, over a realtime spoken channel. Closing this gap opens the door to emerging application areas such as spoken classroom assistants, accessibility-focused tutors for low-literacy learners, multi-party language-learning environments,

and healthcare communication training.

3 System Design and Architecture

[MARKING: This section is worth 10 marks. The marker will check: (1) that each system component is clearly described and its inclusion is well motivated, and (2) that the overall design is coherent — i.e. the components fit together logically. Include an architecture diagram. Explain why *you chose this design, not just* what *it is.]*

3.1 Architecture Overview

hello **?**

[Provide a high-level overview of the full system. Reference your architecture diagram (Figure 1). Explain how data flows through the system from input to output.]

3.2 Component A — [Name]

[Describe this component's role in the system, its inputs and outputs, and why it was chosen over alternatives.]

3.3 Component B — [Name]

3.4 Component C — [Name]

3.5 Design Decisions and Justifications

[Summarise the key design trade-offs you made and why. This helps demonstrate that the overall design is coherent and well-motivated rather than arbitrary.]

4 Experimental Design

4.1 Study Setup

We ran a between-subjects experiment with two conditions. In the first, participants attended a spoken lesson delivered by the AI teacher while an AI peer student was also present. In the second, participants attended the same lesson with the teacher only. Both conditions covered the same topic: Simpson's Paradox. Participants were recruited informally among students at Heriot-Watt University, and each session lasted roughly 10 to 15 minutes. We collected 19 complete responses in total: 9 in the Teacher and AI peer group, and 10 in the Teacher only group.

4.2 Questionnaire Design

After their session, participants filled in a questionnaire with four parts. The first asked them to rate their prior familiarity with Simpson's Paradox on a 1 to 5 scale. The second collected Likert ratings on content clarity, engagement, pace, and post-session confidence. Participants in the Teacher and AI peer group also rated the peer student directly on helpfulness and classroom feel. The third part contained seven knowledge questions, ranging from basic recall to applied reasoning, plus two visual questions. The fourth part gathered open-ended qualitative feedback.

4.3 Analysis

We report group means and run independent-samples t-tests to compare the two conditions. Given the small sample sizes, no result should be treated as conclusive, and the t-tests are included primarily for completeness.

5 Results

5.1 Prior Knowledge

Both groups started from a similar baseline: mean prior familiarity of 1.78 for the Teacher and AI peer group and 1.60 for the Teacher only group, confirming that most participants had little to no prior knowledge of Simpson's Paradox.

5.2 Learning Experience

Table 1 shows mean ratings and t-test results for both groups across the four learning experience items.

Item	T+AI	T-only	t	p
Content clarity	4.22	4.00	0.506	0.620
Engagement	4.00	3.50	1.269	0.222
Pace	4.33	4.10	0.702	0.492
Post-conf.	4.67	4.30	0.946	0.357

Table 1: Mean learning experience ratings by condition (1–5 scale; T+AI $n = 9$, T-only $n = 10$).

The Teacher and AI peer group rated their experience slightly higher across all four dimensions, with the largest gap in engagement (4.00 versus 3.50) and post-session confidence (4.67 versus 4.30). None of the differences reached statistical significance. Participants in the Teacher and AI peer group rated the peer student's helpfulness at 4.33 on average, and its contribution to classroom feel at 5.00, suggesting they found its presence genuinely useful even where the scores did not diverge significantly from the other group.

5.3 Knowledge Assessment

Table 2 reports correct answer counts per question and overall mean scores.

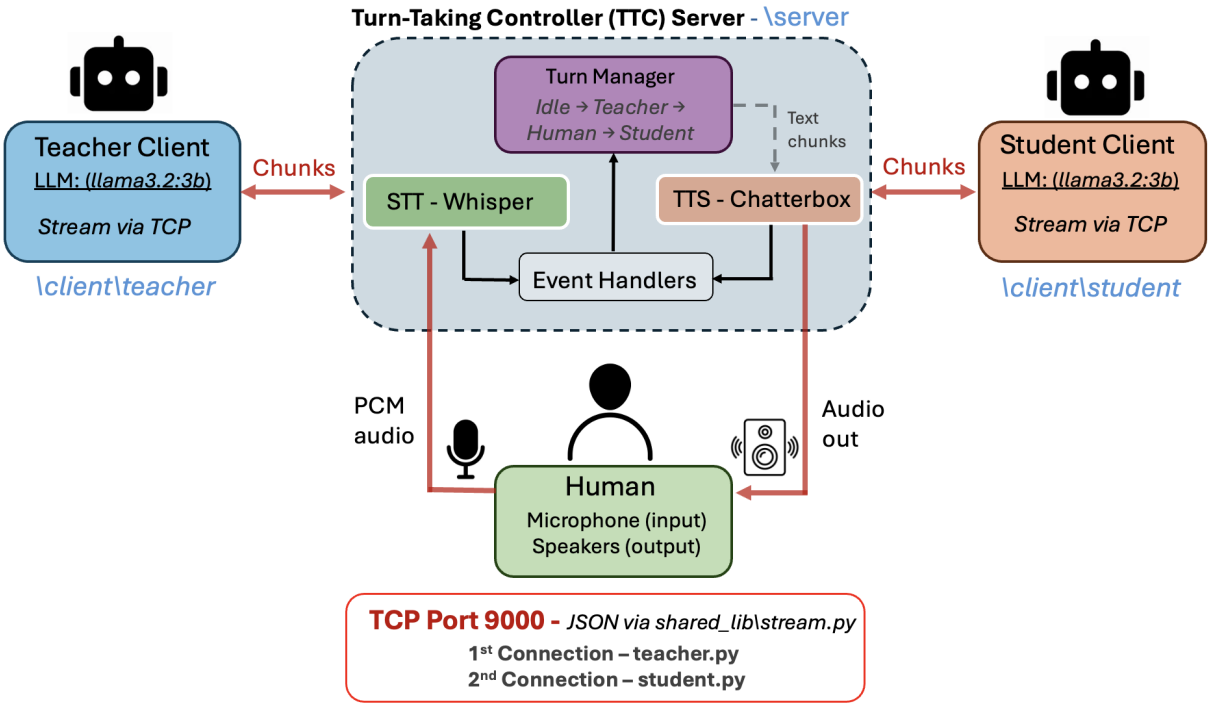

Figure 1: Architecture of the proposed system.

Question	T+AI	T-only
Q9 Definition	8/9	7/10
Q10 Berkeley example	9/9	9/10
Q11 Confounding variables	7/9	7/10
Q12 Common misconception	5/9	4/10
Q13 Critical context	7/9	7/10
Q14 First step	8/9	9/10
Q15 Applied scenario	3/9	5/10
Q16 Visual identification	6/9	7/10
Q17 Practical example	6/9	8/10
Mean /9	**6.67**	**6.30**

Table 2: Knowledge assessment results by condition ($t = 0.382$, $p = 0.707$; T+AI $n = 9$, T-only $n = 10$).

Overall knowledge scores were similar between the two groups (74.1% versus 70.0%), and the t-test found no significant difference (t = 0.382, p = 0.707). Both groups performed well on definitional questions and struggled with the applied scenario (Q15) and the practical example (Q17), where the Teacher only group actually scored slightly higher. Answer confidence followed the same pattern as post-session confidence, with the Teacher and AI peer group rating slightly higher (3.89 versus 3.30, p = 0.190).

5.4 Qualitative Feedback

Responses were broadly positive in both groups. The most consistent criticism was pacing: participants across both conditions found the speech too slow and mentioned that silences between turns broke immersion. Several also noted that the teacher jumped into technical content without a brief introduction, which made the opening harder to follow. In the Teacher and AI peer group, most participants either did not identify the peer as an AI during the session, or said it did not affect their experience when they did. A few noted the peer's questions were helpful. Recurring suggestions included faster TTS, a short contextual introduction before the lesson begins, and more interactive question slots for the human.

5.5 Discussion

The results do not show a clear advantage for either condition in terms of knowledge scores or experience ratings. Both groups reached a similar level of understanding, and no t-test approached significance. The most plausible explanation is that the lesson itself was effective enough to produce consistent results regardless of condition, and that several of the knowledge questions were too straightforward to discriminate between groups. The harder applied questions (Q15, Q17) are more informative in this regard, and the Teacher only group actually performed marginally better on those, though the difference is small and within noise.

Where the two conditions do differ more noticeably is in engagement and post-session confidence, with the Teacher and AI peer group rating both higher. This is consistent with what the vicarious learning hypothesis would predict: having a peer present may en-

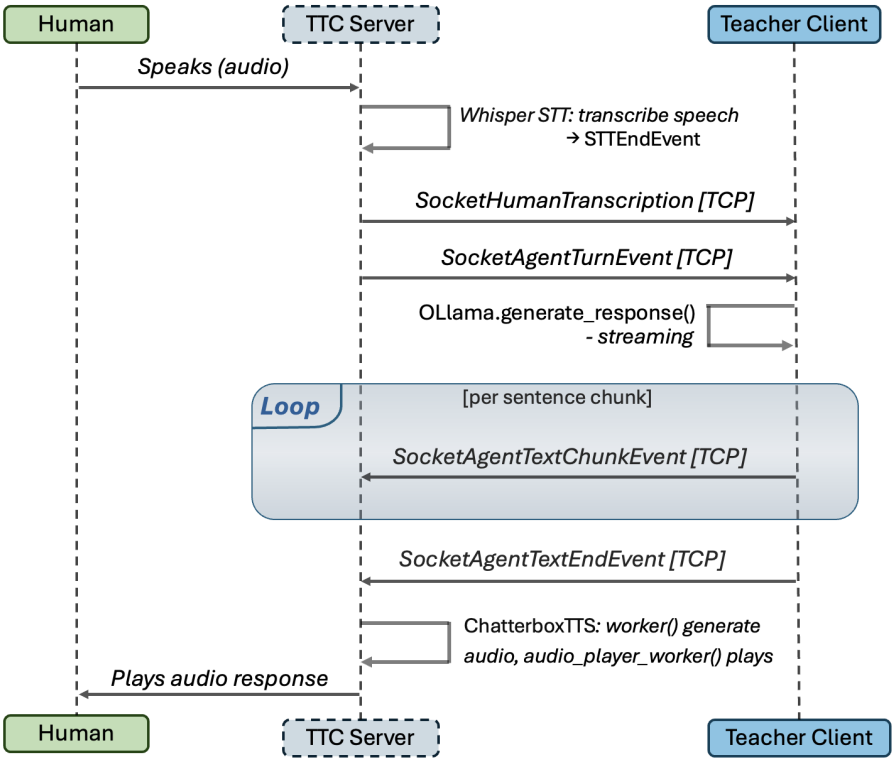

Figure 2: Sequence diagram of the teacher-only interaction workflow (Human, TTC Server, Teacher Client).

rich the perceived experience even when objective test scores are equivalent. The peer student ratings (4.33 for helpfulness, 5.00 for classroom feel) support this reading.

The main limitations are the small and informally recruited sample, the ceiling effect on simpler knowledge questions, and one participant who already knew the system, which biases their responses. It is also worth noting that the conclusions drawn here are specific to an audio-only delivery format. Had the teacher included a visual component such as a slide presentation or a board, more closely replicating a conventional classroom environment, the knowledge scores and engagement ratings may well have diverged more substantially between groups. A larger, more controlled study with pre-screened participants, improved TTS pacing, and a richer visual teaching modality would be needed to draw firmer conclusions.

6 Conclusions

[Summarise the main contributions and findings. Acknowledge limitations honestly. Suggest concrete directions for future research. This section contributes to the overall impression of whether results and conclusions are presented clearly (part of the 20-mark Evaluation criterion).]

[The quality and thoroughness of your references contributes to the Literature Review mark (40 marks). Ensure citations are consistent, complete, and drawn from reputable sources (peer-reviewed papers, established textbooks, official documentation).]

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A Work Assignment

[Describe each team member's or sub-team's responsibilities. This helps the marker understand individual contributions.]

A.1 Team A: [Names]

A.2 Team B: [Names]

A.3 Team C: [Names]

B Questionnaire

[Include the full questionnaire used in extrinsic evaluation so the marker can assess its appropriateness.]

Aspect	Description	Score
Aspect 1	Low (1) – High (5)	
Aspect 2	Low (1) – High (5)	
Aspect 3	Low (1) – High (5)	

Table 3: The questionnaire applied in this report.

C Supplementary Results

[Include any additional tables, statistical tests (e.g. Wilcoxon), or detailed breakdowns that support but would clutter the main text.]

D Ethical Considerations

[Address: (1) informed consent, (2) data protection and GDPR compliance, (3) participant rights (voluntary participation, right to withdraw), (4) risk mitigation, and (5) broader considerations such as bias in conversational agents.]

E Consent Form

F UI Screenshots

[Include screenshots showing typical system behaviours during interaction (e.g. session start, slot filling, handling pauses, self-correction, interruption, session end). These help demonstrate that the system functions well (Implementation criterion, 20 marks).]

Marking Criteria Summary — DELETE BEFORE SUBMISSION

Criterion	Marks	Report Sections
Abstract, Aims, Objectives & Project Description – Abstract succinctly describes project – Research aims are clear	**10**	Abstract, §1
Literature Review – Clarity / readability – Appropriate / thorough references – Good and clear organisation – Critical analysis (MSc only) – Applicability commentary (MSc only)	**40**	§2
System Design and Architecture – Components clear and well motivated – Overall design coherent	**10**	§3
Implementation – Significant amount of work done – System functions well, appropriate complexity	**20**	§4
Evaluation – Methods and metrics discussed thoroughly – Appropriate evaluation plan – Method clear and sensible – Results / conclusions presented clearly	**20**	§5, §6
Total	**100**	

To hide all blue guidance notes, change `\guidancetrue` *to* `\guidancefalse` *in the preamble.*

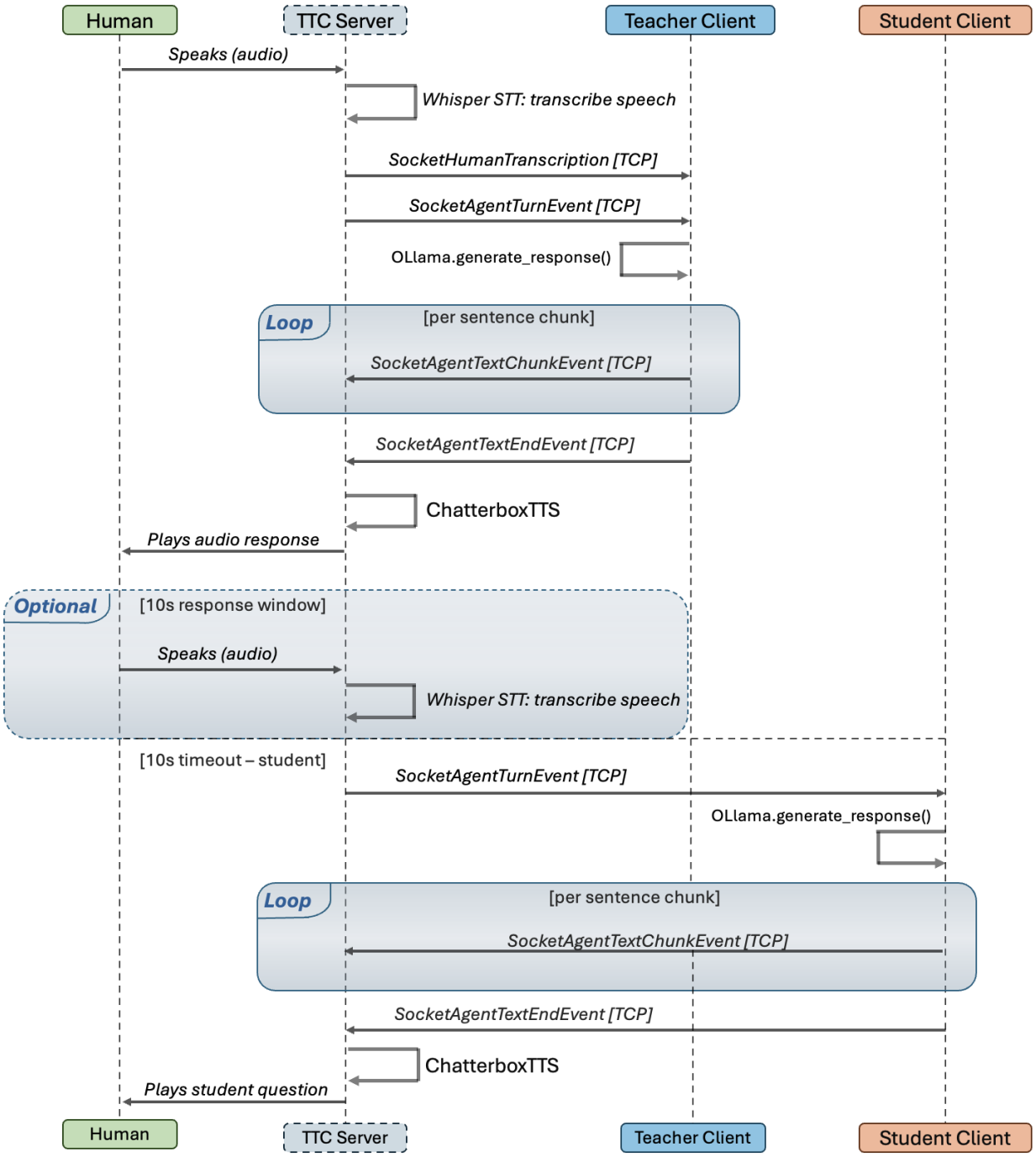

Figure 3: Sequence diagram of the combined teacher–student interaction workflow (Human, TTC Server, Teacher Client, Student Client).